

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 08

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $110 \times 0.67 \times 1.35 \times (0.50 \times 0.90) \times 1.0$ • $= 44.8 \text{ t ha}^{-1} \text{ yr}^{-1}$ (accept 44.7-44.8)	2
01.2	• $110 \times 0.67 \times 1.35 \times (0.35 \times 0.60) \times 0.25$ • $= 5.2 \text{ t ha}^{-1} \text{ yr}^{-1}$	2
01.3	• Beans / up-and-down ploughing: severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall • Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable • Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land ($C = 0.02$) needed to bring loss below $1 \text{ t ha}^{-1} \text{ yr}^{-1}$	3
01.4	• Factors are averages / arbitrary values for typical conditions; rainfall erosivity varies between years / storms • Does not include gully erosion, landslides, wind erosion or local variation in soil; based on data from particular regions	2
01.5	• C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	
02.1	• A fluid in a buried ground loop absorbs low-temperature heat from the ground; at the expansion valve the fluid boils at low pressure absorbing this heat • A compressor raises the pressure so the gas condenses, releasing heat at a higher temperature inside the building • The electricity only moves heat rather than creating it, so up to four times as much heat is delivered as electricity used (less if the source is colder)	3

Question	Marking guidance	Marks
02.2	- Parabolic reflectors focus sunlight onto tubes of oil, increasing the energy density and reaching high temperatures - The heat is transferred to molten salt (potassium nitrate) stored in large insulated tanks at up to 550 °C - The stored heat boils water to drive steam turbines whenever electricity is required, including at night	3
02.3	- Reflectors can only focus parallel rays of direct sunlight; cloud, dust and smoke scatter light so it cannot be concentrated - PV cells convert diffuse as well as direct light, though output is lower	2
02.4	Any 2 from: - Manufacture: extraction / processing of silicon, metals and plastics; toxic wastes (silicon tetrachloride, cadmium) - Land take by large solar farms competing with other uses / affecting desert flora and fauna - Water for cleaning panels in arid areas - Disposal of old panels	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 200-202; AQA spec 3.3.4.1.	
03.1	- Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions - Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected; recirculation increases the metal concentration - Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) - Smelting needs chemical reduction at high temperatures; energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic - Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes	5
03.2	- Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves - The vegetation is harvested and incinerated - The concentrated metal in the ash is dissolved in acid and separated by electrolysis; also decontaminates polluted sites	3
03.3	- Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) - The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
04.1	- Temperature: most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster; but faster decomposition may increase deoxygenation - Light: provides the activation energy for chemical reactions; photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight); light drives photochemical smog reactions producing secondary pollutants - Oxygen: needed for aerobic bacterial decomposition of organic wastes / sewage; oxidation of sulfide ores produces sulfur dioxide / acid mine drainage; oxidation state affects the solubility of metals	6

Question	Marking guidance	Marks
04.2	<i>Any 2 from:</i> • Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment • So they are immobilised and cannot be absorbed by organisms / leached into water; radiation is absorbed in the ground • But they may be released later if conditions change (eg storms disturbing sediment, low pH)	2
04.3	• Many heavy metals (eg lead, aluminium) are more soluble at low pH • So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	
05.1	• $(2900 - 1400) / 1400 \times 100$ • = 107% (accept 107.1)	2
05.2	• Large decrease from 24 per year (1975-79) to 2 per year (2010-14) • Steepest fall between the late 1970s and 1980s; slower decline / levelling off after 1995	2
05.3	<i>Any 3 from:</i> • Double-hulled tankers required for all large tankers • Improved navigation (GPS, AIS) and offshore shipping routes • Inert gas systems reducing explosions • Twin engines / rudders reducing groundings after mechanical failure • Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) • Separate ballast tanks / crude oil washing	3
05.4	• Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff • Routine operations: small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) • Growth in oil use and transport means more small releases even if large accidents are rarer	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
06.1	• Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening • Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight • NO₂ (from morning traffic) is broken down by UV to NO and monatomic oxygen; monatomic oxygen reacts with O₂ to form ozone • Ozone lags behind the NO / NO₂ peaks (which are produced by the morning rush-hour traffic) because the reactions take time; falls at night when there is no sunlight	4
06.2	• NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure • NO is oxidised in the atmosphere (by oxygen / ozone) to NO₂, a secondary pollutant	2
06.3	• PANs (peroxyacetyl nitrates) • Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NO_x)	2

Question	Marking guidance	Marks
06.4	<i>Any 2 from:</i> • Heavy vehicle use giving high NO_x and hydrocarbon emissions • Sunny climate providing the UV for photochemical reactions • Valley / basin topography and frequent temperature inversions trapping pollutants	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
07.1	• Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites	2
07.2	<i>Any 3 from:</i> • Sea current velocity and direction • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people • Geology / hydrology of the coast; sediment deposition areas (mud flats) • Human activities in the area: fishing, seaweed collection, beach use	3
07.3	<i>Any 3 from:</i> • Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) • Monitor their exposure / the concentration of radioactive material in the foods they eat or their body • If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled • CGM assesses exposure before health effects occur (not used to detect damage)	3
07.4	<i>Any 2 from:</i> • Seawater • Sediment / mud • Seaweed • Shellfish / molluscs / fish • Milk, vegetables or meat from nearby farms • Atmospheric dust • Soil	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
08.1	• 1 nitrogen fixation; 2 nitrification • 3 denitrification; 4 ammonification • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three)	3
08.2	• Drainage makes the soil aerobic • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas	3

Question	Marking guidance	Marks
08.3	- Nitrogen gas is very unreactive so most organisms cannot use it directly - Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount	2
08.4	- Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins - When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
09.1	The readings are precise but may not be accurate	1
09.2	- A result that differs from other results with which it was expected to be similar - Repeat the study / measurement: if it recurs it is a 'real' result caused by another variable; if not it was an error and can be excluded	2
09.3	- To test the reliability of the method / equipment - To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable; demand fluctuates daily, weekly, seasonally and short-term; supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants; large-scale electricity storage not yet possible directly so energy must be converted • Pumped-storage HEP: proven, rapid response, peak shaving, but limited sites • Batteries: lithium-ion for grid and vehicles; factors affecting viability; V2G • Hydrogen: electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid; high energy density enables vehicle use • Compressed air with heat recovery; thermal storage (molten salt for CSP, high-volume stores, thermal mass); flywheels; supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage; energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage; judgement	9
10.2	• Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25